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import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler, LabelEncoder

import seaborn as sns

import matplotlib.pyplot as plt


# Load dataset

file_path =

r"C:\\Users\\brian\\OneDrive\\Escritorio\\Lambton\\Semester2\\Toxicology\\Datasets\\PubChem_com
pound_cache.csv"

data = pd.read_csv(file_path)


# Step 1: Handle Missing Values

data.fillna(data.median(numeric_only=True), inplace=True) # Fill numeric columns with median
data.fillna("Unknown", inplace=True) # Fill non-numeric columns with 'Unknown'


# Step 2: Drop Irrelevant Columns

columns_to_drop = ['cmpdsynonym', 'iupacname', 'meshheadings', 'annotation']

data.drop(columns=columns_to_drop, inplace=True, errors='ignore')


# Step 3: Label Encode Categorical Variables

categorical_columns = data.select_dtypes(include=['object']).columns

for col in categorical_columns:

    data[col] = LabelEncoder().fit_transform(data[col])


# Step 4: Select Features and Target

# Replace 'annotation' with your actual target column if applicable
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```
target_column = 'mw' # Example: use molecular weight for testing (update with your target variable)

if target_column not in data.columns:

    raise ValueError("Specify a valid target column.")

X = data.drop(columns=[target_column])

y = data[target_column]


#####3

# Preparing features and target for the first model (e.g., is_toxic classification)

data['is_toxic'] = (data['hbondddonor'] > 0).astype(int) # Example logic for toxicity


# Define features and target for Model 1

features_model_1 = ['mw', 'polararea', 'complexity', 'xlogp', 'heavycnt',

                    'hbondddonor', 'rotbonds', 'totalatomstereocnt', 'totalbondstereocnt']

target_model_1 = 'is_toxic' # Update this if your target column name is different


# Prepare data for Model 1

X1 = data[features_model_1]

y1 = data[target_model_1]


# Train/Test Split

from sklearn.model_selection import train_test_split

X1_train, X1_test, y1_train, y1_test = train_test_split(X1, y1, test_size=0.2, random_state=42)


# Train RandomForestClassifier

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score
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```
model_1 = RandomForestClassifier(random_state=42)
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```
model_1.fit(X1_train, y1_train)
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# Evaluate Model 1
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y1_pred = model_1.predict(X1_test)
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print(f"Model 1 Accuracy: {accuracy_score(y1_test, y1_pred)}")
```

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# Save Model 1
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```
import os
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```
import joblib
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output_dir = r"C:\\Users\\brian\\OneDrive\\Escritorio\\Lambton\\Semester2\\Toxicology"
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joblib.dump(model_1, os.path.join(output_dir, 'model_1_toxicity.pkl'))
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